

G - Children Yearn for the Evil Kindergarten

分值: 625 分

Problem Statement

There are 10^{100} kids at a game venue. Initially, no kid has any medals.

游戏场馆里共有 10^{100} 个孩子。初始时，所有孩子都没有奖牌。

A kid leaves the venue exactly when they either **drop out** or **escape**.

当且仅当孩子**被淘汰**或**逃离**时，他们会离开场馆。

The game consists of N days. On day i ($1 \leq i \leq N$), the following sequence of operations is performed in order.

游戏持续 N 天。在第 i 天 ($1 \leq i \leq N$)，将按顺序执行以下操作序列。

- Collect all medals held by the kids at the venue, and let s be the total number of collected medals.
- Distribute $s + A_i$ medals freely among the kids at the venue (if there are no kids at the venue, do nothing).
- Among the kids at the venue, those with fewer than B_i medals drop out. Those with at least B_i medals each lose B_i medals.
- Among the kids at the venue, those with at least C_i medals each choose whether to escape at this point or remain at the venue.
- 收回场馆内所有孩子持有的奖牌，记收回的奖牌总数为 s 。
- 将 $s + A_i$ 枚奖牌自由分配给场馆内的孩子（如果场馆内没有孩子则不执行操作）。
- 在场馆内的孩子中，奖牌数少于 B_i 的孩子将被淘汰；奖牌数不少于 B_i 的孩子每人扣除 B_i 枚奖牌。
- 在场馆内的孩子中，奖牌数不少于 C_i 的孩子可以选择此时逃离，或者留在场馆内。

Kids who remain at the venue at the end of the N days drop out.

在 N 天结束后仍留在场馆内的孩子将被淘汰。

Find the maximum possible number of kids who ultimately escape.

求最终能够逃离的孩子的最大可能数量。

You are given T test cases; solve each of them.

给定 T 组测试数据，你需要解决每组数据。

Constraints

- $1 \leq T \leq 3 \times 10^5$
 - $1 \leq N \leq 3 \times 10^5$
 - $1 \leq A_i \leq 10^6$
 - $1 \leq B_i \leq 10^6$
 - $1 \leq C_i \leq 10^6$
 - The sum of N over all test cases is at most 3×10^5 .
 - All input values are integers.
 - $1 \leq T \leq 3 \times 10^5$
 - $1 \leq N \leq 3 \times 10^5$
 - $1 \leq A_i \leq 10^6$
 - $1 \leq B_i \leq 10^6$
 - $1 \leq C_i \leq 10^6$
 - 所有测试数据的 N 之和不超过 3×10^5 。
 - 所有输入值均为整数。
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Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

T
case₁
case₂
⋮
case _{T}

Each test case is given in the following format:

每组测试数据按照以下格式给出：

N
 A_1 B_1 C_1
 A_2 B_2 C_2
⋮
 A_N B_N C_N

Output

Output the answers for the test cases in order, separated by newlines.

按顺序输出每组测试数据的答案，答案之间用换行分隔。

Sample Input 1

```
2
4
16 2 3
15 2 4
1 3 5
20 5 5
2
41404 1 941738
211877 205711 417821
```

Sample Output 1

```
5
0
```

Consider the first test case. By acting as follows, five kids can escape.

考虑第一组测试数据。通过如下操作，可以让5个孩子逃离。

- At the start of day 1, collect $s = 0$ medals from 10^{100} kids. Then, proceed as follows.
 - Distribute $0 + 16 = 16$ medals so that the kids' medal counts become $(5, 5, 2, 2, 2, 0, \dots, 0)$.
 - The $10^{100} - 5$ kids with no medals drop out, and the remaining 5 kids' medal counts become $(3, 3, 0, 0, 0)$.
 - The 2 kids with 3 medals each choose to escape, and the remaining 3 kids' medal counts become $(0, 0, 0)$.
 - At the start of day 2, collect $s = 0$ medals from 3 kids. Then, proceed as follows.
 - Distribute $0 + 15 = 15$ medals so that the kids' medal counts becomes $(6, 6, 3)$.
 - No one drops out, and the remaining 3 kids' medal counts become $(4, 4, 1)$.
 - 1 kid with 4 medals chooses to escape, and the remaining 2 kids' medal counts become $(4, 1)$.
 - At the start of day 3, collect $s = 5$ medals from 2 kids. Then, proceed as follows.
 - Distribute $5 + 1 = 6$ medals so that the kids' medal counts becomes $(3, 3)$.
 - No one drops out, and the remaining 2 kids' medal counts become $(0, 0)$.
 - No one escapes.
 - At the start of day 4, collect $s = 0$ medals from 2 kids. Then, proceed as follows.
 - Distribute $0 + 20 = 20$ medals so that the kids' medal counts becomes $(10, 10)$.
 - No one drops out, and the remaining 2 kids' medal counts become $(5, 5)$.
 - The 2 kids with 5 medals each choose to escape, and the venue becomes empty.
 - 在第 1 天开始时，从 10^{100} 个孩子处收回 $s = 0$ 枚奖牌。然后执行如下操作：
 - 分配 $0 + 16 = 16$ 枚奖牌，使得孩子们的奖牌数变为 $(5, 5, 2, 2, 2, 0, \dots, 0)$ 。
 - $10^{100} - 5$ 个没有奖牌的孩子被淘汰，剩余 5 个孩子的奖牌数变为 $(3, 3, 0, 0, 0)$ 。
 - 拥有 3 枚奖牌的 2 个孩子选择逃离，剩余 3 个孩子的奖牌数变为 $(0, 0, 0)$ 。
 - 在第 2 天开始时，从 3 个孩子处收回 $s = 0$ 枚奖牌。然后执行如下操作：
 - 分配 $0 + 15 = 15$ 枚奖牌，使得孩子们的奖牌数变为 $(6, 6, 3)$ 。
 - 没有人被淘汰，剩余 3 个孩子的奖牌数变为 $(4, 4, 1)$ 。
 - 拥有 4 枚奖牌的 1 个孩子选择逃离，剩余 2 个孩子的奖牌数变为 $(4, 1)$ 。
 - 在第 3 天开始时，从 2 个孩子处收回 $s = 5$ 枚奖牌。然后执行如下操作：
 - 分配 $5 + 1 = 6$ 枚奖牌，使得孩子们的奖牌数变为 $(3, 3)$ 。
 - 没有人被淘汰，剩余 2 个孩子的奖牌数变为 $(0, 0)$ 。
 - 没有人逃离。
 - 在第 4 天开始时，从 2 个孩子处收回 $s = 0$ 枚奖牌。然后执行如下操作：
 - 分配 $0 + 20 = 20$ 枚奖牌，使得孩子们的奖牌数变为 $(10, 10)$ 。
 - 没有人被淘汰，剩余 2 个孩子的奖牌数变为 $(5, 5)$ 。
 - 拥有 5 枚奖牌的 2 个孩子选择逃离，场馆变为空。
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In the second test case, not a single kid can escape.

在第二组测试数据中，没有孩子能够逃离。

